

Xiaohui Cai

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SUMMARY

Backend engineer with 7 years across Go, Python and Java, now building AI infrastructure for a computer-vision SaaS serving 45+ brand clients. Depth in multi-cluster Kubernetes, observability from scratch, gRPC service governance and GPU inference serving. **Nacos Committer** — 135 merged PRs in the alibaba/nacos ecosystem.

EXPERIENCE

Senior Backend Engineer — LenzTech

Jul 2025 – Present

Retail shelf-recognition SaaS. Go / Python / Java microservices on multi-cluster Kubernetes, ~1M image recognition requests per day, GPU inference on qGPU-shared nodes.

- Built the observability stack from zero to production — Prometheus + Alertmanager covering queue backlog, OOM, CPU throttling, GPU memory and rate-limit queueing, every rule live-fire tested before rollout.
- Rolled out OpenTelemetry + Jaeger across all services plus node-level timing in the Go DAG engine, which attributed a 26s latency alert to 24.5s of rate-limit queueing versus 1.4s of real execution.
- Designed and led a unified NVIDIA Triton inference platform — deployment lifecycle, APISIX gRPC header routing, versioning, Redis concurrency locks — consolidating 90 inference services that implemented only 36 distinct functions.
- Root-caused production incidents: a Redis FLUSHDB that broke recognition platform-wide (found via slowlog), a 7-day Kafka consumer stall that silently dropped 1,157 records, and a gRPC single-worker deadlock.
- Shipped release safeguards — consume-pause during model rollout (zero message loss) and graceful shutdown draining 263s-average requests — and cut cluster logs from 60 to 10–15 GB/day, hot-reloadable via Nacos.
- Built a multi-agent GitLab code review service and a shared Claude Code plugin marketplace with 10+ team skills.

Senior Java Engineer — InsightOne

Apr 2022 – Jul 2025

- Owned private information retrieval (PIR) and certificate management on a financial-grade privacy-preserving computation platform (secure multi-party computation, federated learning): integrated the native cryptographic libraries, shipped the PIR upgrade, and stood up a private CA with mutual TLS between nodes.
- Delivered 10+ on-premise deployments for power grid, securities exchange, taxation, banking, telecom research and logistics clients — on-site setup, network troubleshooting and customer-side blockchain integration.
- Ported the full stack to domestic Chinese infrastructure: OceanBase, Hygon / Kunpeng (x86 + ARM), Kylin OS, TongWeb / TongRDS, including an ARM + IPv6 deployment.

Java Engineer — Ruiyou Technology, onsite at JD Logistics

Aug 2019 – Apr 2022

- Built collateral valuation and pledge management for the inventory-financing product on JD's supply-chain finance platform, and integrated cross-border financing with DBS Bank Singapore.
- Led Singles' Day readiness: read/write splitting and index tuning to relieve primary-database CPU saturation, removed an in-loop database connection (12s → under 1s), and cleared queue backlog with InnoDB row-lock indexing plus ordered messaging. Peaked at 18,500 TPM with no incidents.

OPEN SOURCE

Nacos Committer — alibaba/nacos ecosystem

Feb 2024 – Present

Alibaba's open-source service discovery and configuration platform (30k+ stars). 135 merged PRs across 10+ repositories over 2.5 years; promoted to Committer in May 2026.

- Led the gRPC infrastructure rewrite for nacos-sdk-go v3 — PayloadCodec layer, auth token refresh, proto migration.
- Bootstrapped nacos-sdk-proto, the multi-language proto ecosystem, with its CI/CD pipeline and version compatibility policy.
- Drove ecosystem-wide code quality (SpotBugs gating + Spotless, 14 modules); patched CVE-2024-22243 / 22259 / 22262.

TECHNICAL SKILLS

Languages & Frameworks Go, Python, Java, TypeScript; Spring Boot, Spring Cloud Alibaba, FastAPI, Vue3

Cloud Native Kubernetes, Docker, Helm, ArgoCD, APISIX, gRPC, Nacos, Kafka; Alibaba Cloud (ACK, VPC, SLB/ALB, OSS)

Observability Prometheus, Alertmanager, Grafana, OpenTelemetry, Jaeger

Data & AI Infrastructure MySQL, MongoDB, Redis; NVIDIA Triton Inference Server, qGPU scheduling, ONNX Runtime

EDUCATION

Inner Mongolia University of Technology — B.S., Computer Science and Technology

Mar 2021 – Jun 2023